LAB: 02

Overview of hashing, file header information, disk images

Objective:
- **Learn about Data integrity, File Headers, view metadata and File Manipulation**

Activities:
Task 1
File Hashes

Leaning Activities:

At the end of these activities, you should understand:
- How to calculate and compare hashes of the files
- How detect NTFS stream data
- How to display contents in steganography

Tools to use:
-Windows 10 Host (VM or host PC doesn't matter)
-WinHex (available from: http://www.x-ways.net/winhex.zip)
-HxD Hex Editor (available from: https://mh-nexus.de/en/downloads.php?product=HxD20)
-Nirsoft Hashmyfiles (available from: https://www.nirsoft.net/utils/hashmyfiles-x64.zip)

Note: Please include a screenshot of each step. Corp your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

Sample files contained within Lab2.zip

Task 1:

Hashes
A Hash (MD5, Sha1, etc) also known as 'a Cryptographic hash function' is a "One-Way" function which describes the applicable data in a numeric string of a fixed length. This is often much smaller than the original data. By design a Hash function is not feasible to reverse to derive the described original data.

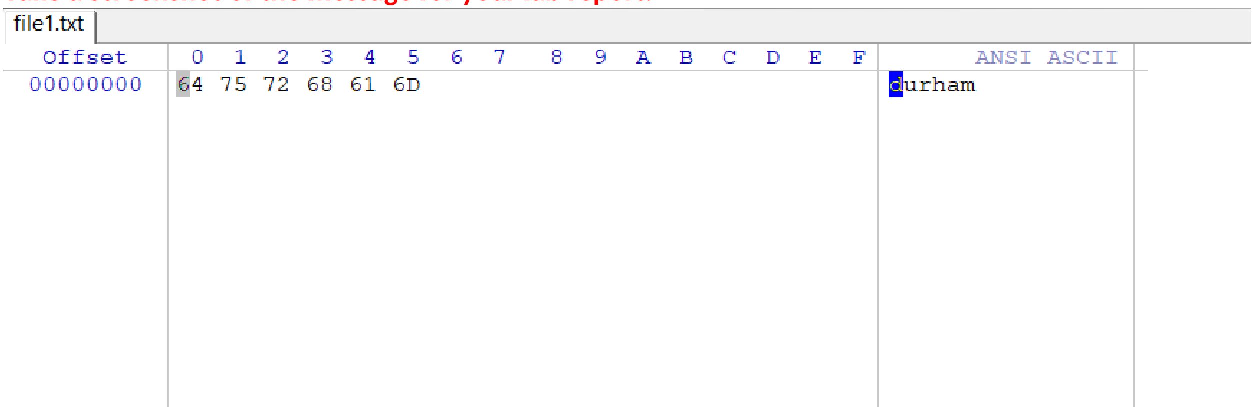
Diffusion property of hashes: if you change a small amount of the original data (1 bit) due to the construction of the hashing algorithm a large change occurs to the hashing output.

1.1 Open notepad and type the following text:

It just says a single word 'Durham'.

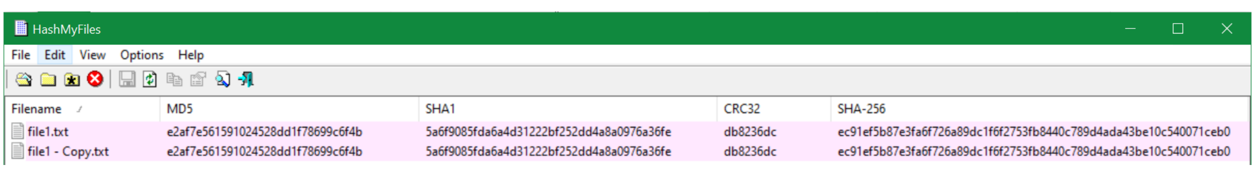
1.2 Save as file1.txt

1.3 Open file1.txt in Winhex and view the contents
In WinHex click 'file' then 'open' then select file1.txt. On the left is the data contained within the file in hex code, on the right is the ANSI ASCII translation of that hex code.
Take a screenshot of the message for your lab report.

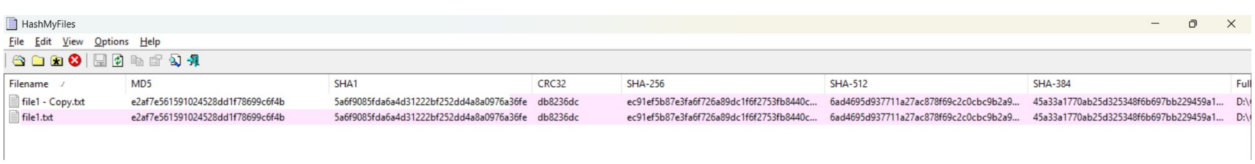


1.4 Now, let's copy file1.txt and paste it. We should now have file1.txt and file1 – Copy.txt in our folder.

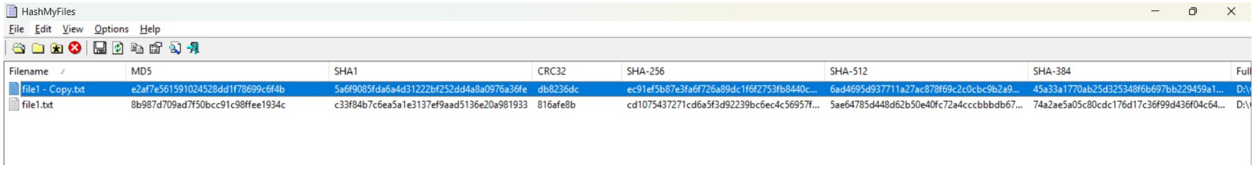
1.5 Let's use the hash tool (Nirsoft Hashmyfiles) to calculate the hash of both of these files.
At this point the hash should be the same for both files (exact copies of one-another).



In fact, your file's hash should match mine. Hashes are deterministic (Figure 1.5).
Take a screenshot of the message for your lab report.



1.6 Now open file1.txt in notepad and add a period, save, and rehash the file and compare it to our copy of the original. Notice the difference in the resulting hash value.
Take a screenshot of the message for your lab report.



Task 2:
A txt file is very simple and doesn't contain any header detail aside from the content of the text file (Durham). However, most files have a lot more information and details baked in.

2.1 Let's take for instance file2.rtf this file is called a rich text file. The file has a lot more features than a simple txt file and since that's the case, more information is contained within it. Let's look at file2.rtf in WinHex.

More complex files often have more functionality, in this case the rtf file contains information regarding the stored data such as the keyset, font, and other control verbs to help the rtf reader render the document correctly with the correct format.

2.2 Let's open up both file2.rtf and file 3.rtf in Wordpad and look at the contents.
Both files say 'durham'.

2.3 Now let's run a hash against these files, they appear to be copies of one-another, so do their hashes match?
Why or Why don't the hashes match? (include your answer in your lab write up)

No, the hashes do not match. That is because the metadata is different of file 2 and file 3.

To be precise, the creation timestamp on each file is different.

As entirety of file is hashed metadata is included which includes creation timestamps.

	[unregistered]
file2.rtf	
D:\College\Sem 2 (Cybersecurity)\	
File size:	176 B
	176 bytes
Default Edit Mode	
State:	original
Undo level:	0
Undo reverses:	n/a
Creation time:	2020-01-14
	14:54:20
Last write time:	2025-05-24
	18:48:43
Attributes:	A
Icons:	0
Mode:	Text
Offsets:	hexadecimal
Bytes per page:	46x16=736

File 2.rtf metadata

	[unregistered]
file 3.rtf	
D:\College\Sem 2 (Cybersecurity)\	
File size:	176 B
	176 bytes
Default Edit Mode	
State:	original
Undo level:	0
Undo reverses:	n/a
Creation time:	2020-01-14
	15:15:20
Last write time:	2025-05-24
	18:48:43
Attributes:	A
Icons:	0
Mode:	Text
Offsets:	hexadecimal
Bytes per page:	46x16=736

File 3.rtf metadata

2.4 Let's use WinHex or HxD Editor to look a bit deeper into these two files.
Use the Tools/Compare data feature in WinHex or HxD to determine what's going on with these two files.

2.5 Something is strange on offset (the number of bytes from the beginning of the file) 173 what is it?

Yes, the hex numbers are different but only for one byte at offset 0x90. File 2.rtf has 0D while file 3.rtf Has 1D.

2.6 Let's open file 4.rtf in wordpad.
We see it says durham, again, looks the same as file 3 and file 2. However, this file is hiding a dark secret!! Open the file in WinHex to see a hidden message.
Take a screenshot of the message for your lab report.

File2.rtf

```
file2.rtf | file 3.rtf | file 4.rtf

Offset    0  1  2  3  4  5  6  7   8  9  A  B  C  D  E  F   ANSI ASCII
00000000  7B 5C 72 74 66 31 5C 61  6E 73 69 5C 64 65 66 66  {\rtf1\ansi\deff
00000010  30 5C 6E 6F 75 69 63 6F  6D 70 61 74 7B 5C 66 6F  0\nouicompat{\fo
00000020  6E 74 74 62 6C 7B 5C 66  30 5C 66 6E 69 6C 5C 66  nttbl{\f0\fnil\f
00000030  63 68 61 72 73 65 74 30  20 43 61 6C 69 62 72 69  charset0 Calibri
00000040  3B 7D 7D 0D 0A 7B 5C 2A  5C 67 65 6E 65 72 61 74  ;}}  {\*\generat
00000050  6F 72 20 52 69 63 68 65  64 32 30 20 31 30 2E 30  or Riched20 10.0
00000060  2E 31 38 33 36 32 7D 5C  76 69 65 77 6B 69 6E 64  .18362}\viewkind
00000070  34 5C 75 63 31 20 0D 0A  5C 70 61 72 64 5C 73 61  4\uc1   \pard\sa
00000080  32 30 30 5C 73 6C 32 37  36 5C 73 6C 6D 75 6C 74  200\sl276\slmult
00000090  31 5C 66 30 5C 66 73 32  32 5C 6C 61 6E 67 39 20  1\f0\fs22\lang9
000000A0  64 75 72 68 61 6D 5C 70  61 72 0D 0A 7D 0D 0A 00  durham\par   }
```

File3.rtf

```
file2.rtf | file 3.rtf | file 4.rtf

Offset    0  1  2  3  4  5  6  7   8  9  A  B  C  D  E  F   ANSI ASCII
00000000  7B 5C 72 74 66 31 5C 61  6E 73 69 5C 64 65 66 66  {\rtf1\ansi\deff
00000010  30 5C 6E 6F 75 69 63 6F  6D 70 61 74 7B 5C 66 6F  0\nouicompat{\fo
00000020  6E 74 74 62 6C 7B 5C 66  30 5C 66 6E 69 6C 5C 66  nttbl{\f0\fnil\f
00000030  63 68 61 72 73 65 74 30  20 43 61 6C 69 62 72 69  charset0 Calibri
00000040  3B 7D 7D 0D 0A 7B 5C 2A  5C 67 65 6E 65 72 61 74  ;}}  {\*\generat
00000050  6F 72 20 52 69 63 68 65  64 32 30 20 31 30 2E 30  or Riched20 10.0
00000060  2E 31 38 33 36 32 7D 5C  76 69 65 77 6B 69 6E 64  .18362}\viewkind
00000070  34 5C 75 63 31 20 0D 0A  5C 70 61 72 64 5C 73 61  4\uc1   \pard\sa
00000080  32 30 30 5C 73 6C 32 37  36 5C 73 6C 6D 75 6C 74  200\sl276\slmult
00000090  31 5C 66 30 5C 66 73 32  32 5C 6C 61 6E 67 39 20  1\f0\fs22\lang9
000000A0  64 75 72 68 61 6D 5C 70  61 72 0D 0A 7D 1D 0A 00  durham\par   }
```

File4.rtf

file2.rtf | file 3.rtf | file 4.rtf

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	ANSI ASCII
00000000	7B	5C	72	74	66	31	5C	61	6E	73	69	5C	61	6E	73	69	{\rtf1\ansi\ansi
00000010	63	70	67	31	32	35	32	5C	64	65	66	66	30	5C	6E	6F	cpg1252\deff0\no
00000020	75	69	63	6F	6D	70	61	74	7B	5C	66	6F	6E	74	74	62	uicompat{\fonttb
00000030	6C	7B	5C	66	30	5C	66	6E	69	6C	5C	66	63	68	61	72	l{\f0\fnil\fchar
00000040	73	65	74	30	20	43	61	6C	69	62	72	69	3B	7D	7D	0D	set0 Calibri;}}
00000050	0A	7B	5C	2A	5C	67	65	6E	65	72	61	74	6F	72	20	52	{*\generator R
00000060	69	63	68	65	64	32	30	20	31	30	2E	30	2E	31	38	33	iched20 10.0.183
00000070	36	32	7D	5C	76	69	65	77	6B	69	6E	64	34	5C	75	63	62}\viewkind4\uc
00000080	31	20	0D	0A	5C	70	61	72	64	5C	73	61	32	30	30	5C	1 \pard\sa200\
00000090	73	6C	32	37	36	5C	73	6C	6D	75	6C	74	31	5C	66	30	sl276\slmult1\f0
000000A0	5C	66	73	32	32	5C	6C	61	6E	67	39	20	64	75	72	68	\fs22\lang9 durh
000000B0	61	6D	5C	70	61	72	7D	54	68	69	73	20	69	73	20	68	am\par}This is h
000000C0	69	64	64	65	6E	20	66	72	6F	6D	20	76	69	65	77	2E	idden from view.
000000D0	20	20	20	20	20	20	20	20	0D	0A	00						

This is a form a steganography, that is, hiding information or a message within another file.

2.7 Why would steganography be important to identify in digital forensics?

As an attacker steganography is very important because it can be used to inject viruses, malwares or even commands that work in background. This is done by embedding code to any file for example image, pdf or in this case text file. Checking for steganography is important in digital forensics because it can reveal important information which is not visible to eyes or while opening file normally.

From digital forensics perspective, criminals and cyber criminals hides data such as malicious code or secret message through steganography so learning and detecting such activities can give many evidences and leads to catch them or to present evidence in court.

Task 3

1. Download s-tools (http://www.jjtc.com/Steganography/tools.html)
   Save the zip file in C:\task3 folder.

2. Unzip the zip file in C:\task3\steg folder.

3. Drag the zebras.bmp file to your desktop. Do not make a shortcut. The file itself must be moved there.

4. Start S-tools.exe by double clicking on the icon on the desktop. A window will appear.

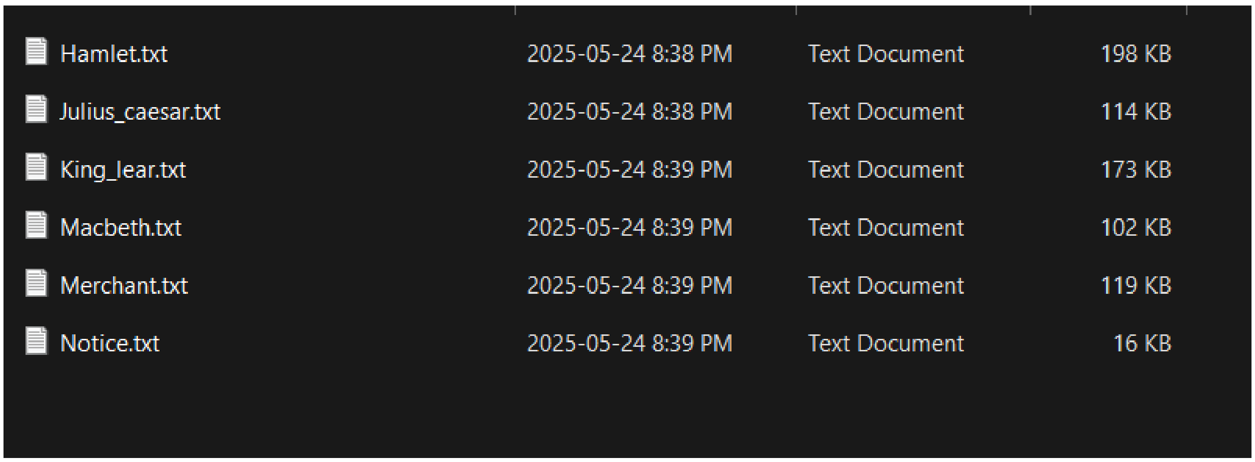
5. Drag the zebras.bmp file to the S-tools window.

6. Right click on the zebras pictures and select Reveal from the menu.

7. Fill in the 3-character pass phrase 'abc' (without the quotes) in two places. Leave IDEA as the encryption algorithm. Click on OK.

8. Wait until the Revealed Archive dialog box appears. This may take a minute or two.

9. Right click on any item and select Save As to save the file. Repeat for the other ones. These are the hidden files. **Take a screenshot of the message for your lab report.**



Hamlet.txt	2025-05-24 8:38 PM	Text Document	198 KB
Julius_caesar.txt	2025-05-24 8:38 PM	Text Document	114 KB
King_lear.txt	2025-05-24 8:39 PM	Text Document	173 KB
Macbeth.txt	2025-05-24 8:39 PM	Text Document	102 KB
Merchant.txt	2025-05-24 8:39 PM	Text Document	119 KB
Notice.txt	2025-05-24 8:39 PM	Text Document	16 KB

Task 4

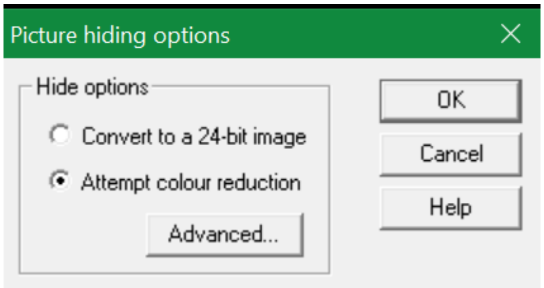
1. Create a s1.txt file and add the following message and save the file:
   The hidden phone#:
   123 455 6666
   443 555 3444
   890 343 3333

2. Launch s-tools

3. Drag task4.bmp file into the open windows

4. Now drag the “s1.txt” over the top of the BMP image. The next window that you see will be asking you to enter the passphrase and to select your encryption algorithm. Let’s leave the algorithm as the default “IDEA” and type the passphrase "123456" in, If you typed correctly you’ll now see a second picture of the BLACK only this time it will have Hidden Data in the top blue bar.

   If the following msg showed up, click OK



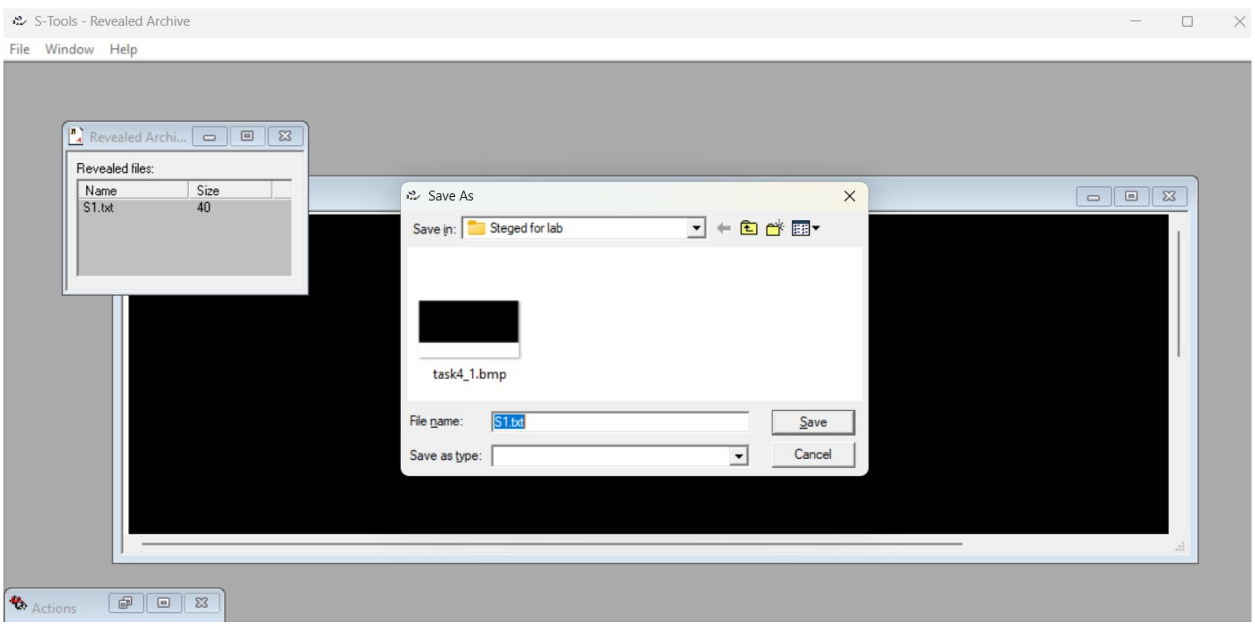
5. Now let's right click that new "hidden data" picture and then Save As and name it "task4_1.bmp"

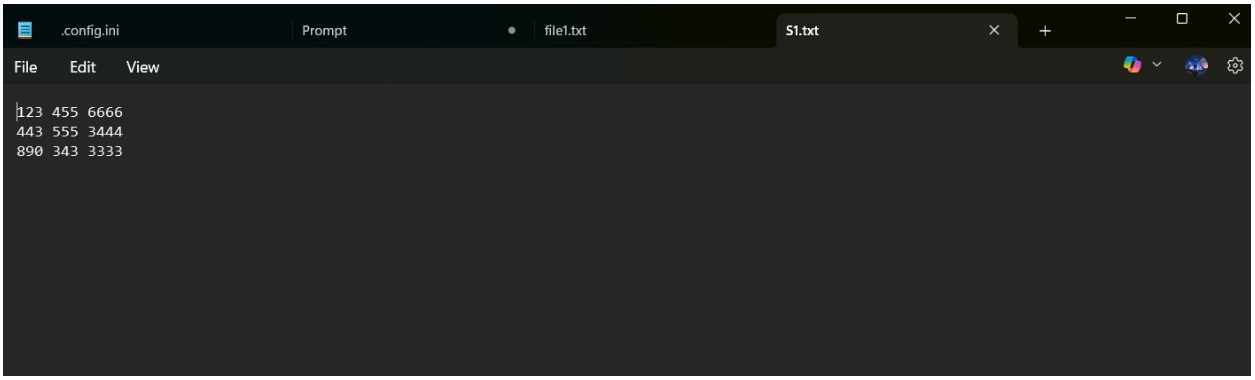
6. Now close out of everything and re-open S-Tools.exe. Next, drag the "task4_1.bmp" file onto the S-Tools window, move your mouse over the picture and right-click. Select the option "Reveal" and you will see the same window you saw before requesting the passphrase. Type in your passphrase, 123456 and click OK.

7. Wait until the Revealed Archive dialog box appears. This may take a minute or two.

8. Right click on any item and select Save As to save the file to desktop.

Take a screenshot of the message for your lab report.





Task5
1.create a new directory c:\task5

2. Go to a task5 directory and create a txt file

3. Right Click, select New > Text document, enter the name "lab2_task5.txt"

4. Open the file in notepad and enter text "This is a demo file for ADS.", Save and close.

5. Open a command prompt Start > Run, enter "cmd", click OK

6. In the command window, go to your task5 directory.

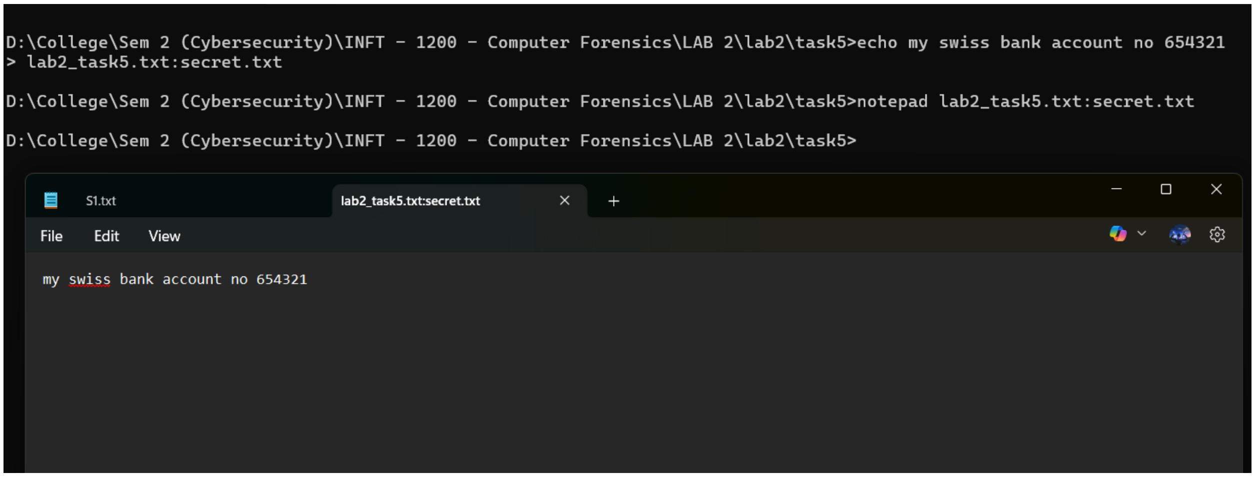
7. Type the following to confirm that you have some text in the file: notepad lab2_task5.txt
A notepad window will pop up and show your lab2_task5.txt text.

This has all been setup. Now we get to the ADS part.

8.Type the following to save secret notes to a different text file and hide the reference to that text file inside of "lab3_task6.txt", echo my swiss bank account no 654321 > lab2_task6.txt:secret.txt

9. Now if you type the following command you will see only your original file:
"notepad lab3_task6.txt"

Try the following command to see the secret message:
"lab3_task6.txt:secret.txt"
Take a screenshot of the message for your lab report.

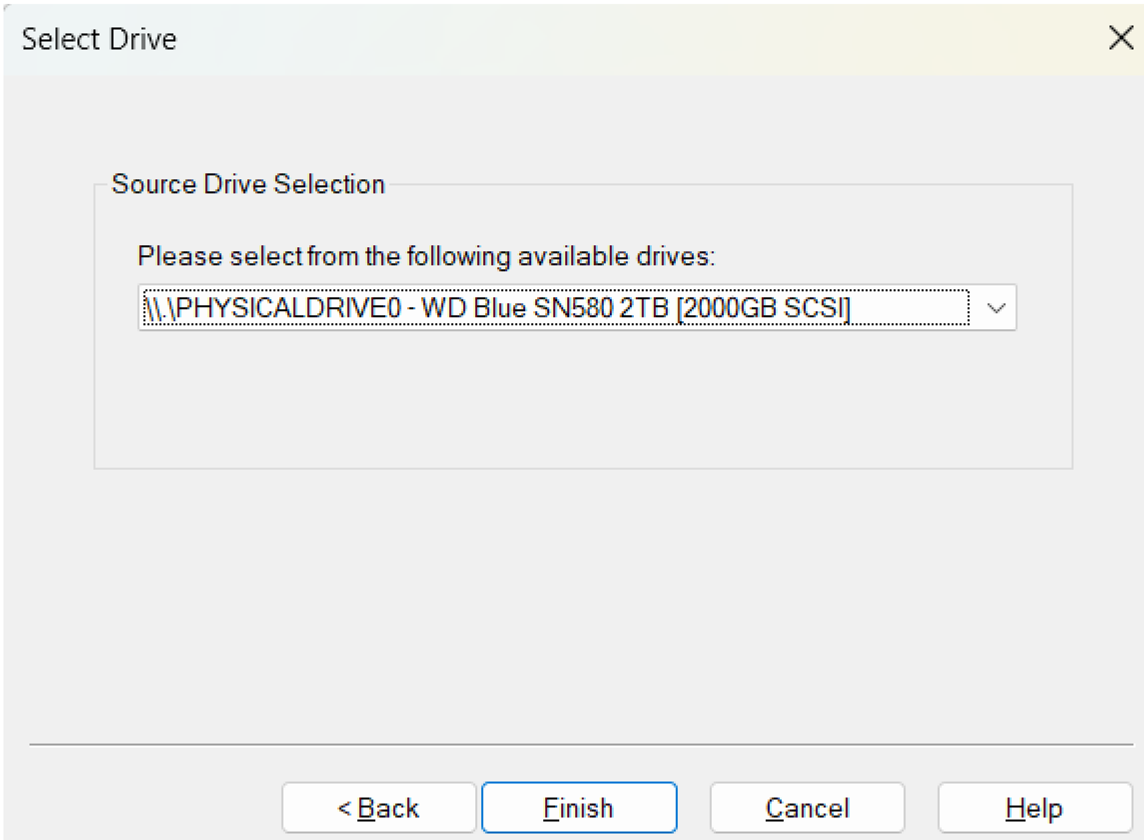


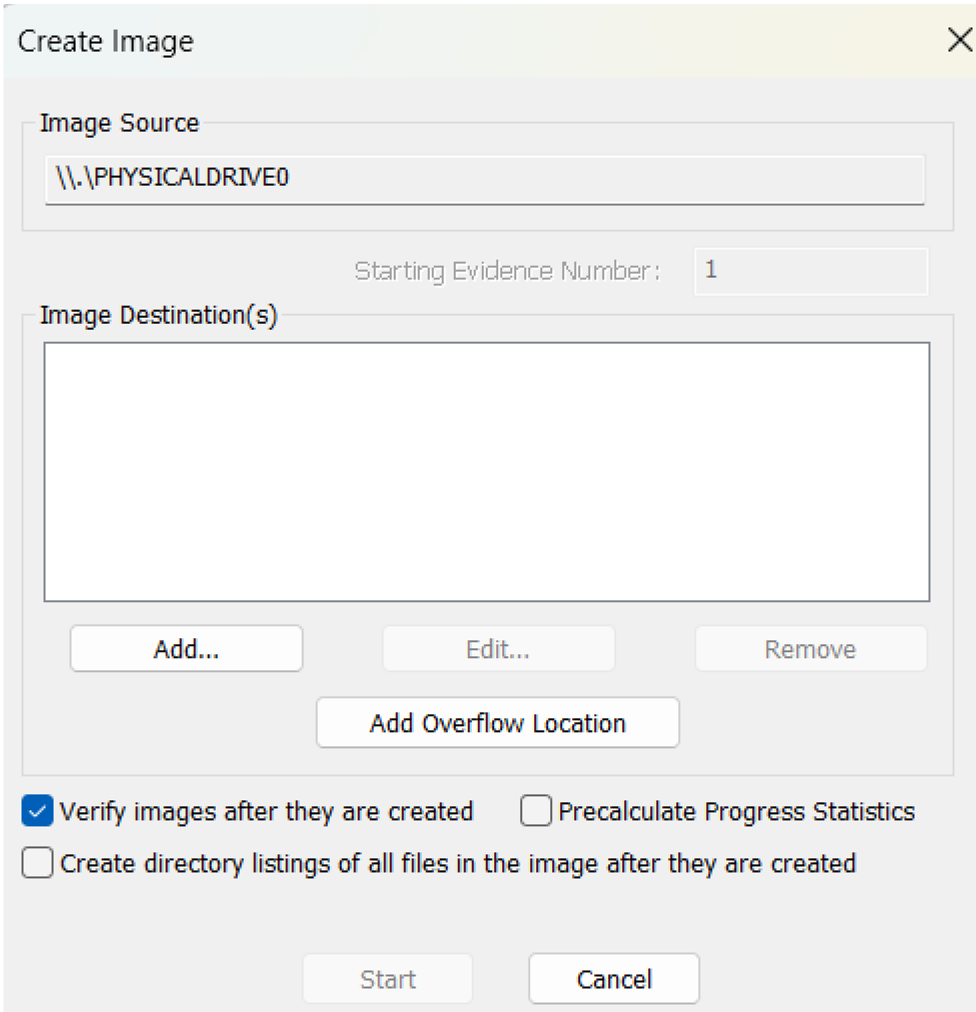
Task6

Text book, Page 150 & 151

Steps 1-7

Take a screenshot of the message for your lab report.


Select Drive
Source Drive Selection
Please select from the following available drives:
\\.\PHYSICALDRIVE0 - WD Blue SN580 2TB [2000GB SCSI]
< Back
Finish
Cancel
Help


Create Image
Image Source
\\.\PHYSICALDRIVE0
Starting Evidence Number:
1
Image Destination(s)
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel

Reflective statements (end-of-exercise):
You should reflect on these questions:

1. Steganalysis is the process of detecting steganography and recovering hidden evidence.

2. What is the difference between cryptography and steganography?

Basic difference of cryptography and steganography is that cryptography encrypts data and steganography hides data itself.

In cryptography message or data of file is visible but encrypted while in steganography data or message is not visible and hidden in a file like image, text, pdf or even in video.

Cryptography is used for integrity and confidentiality however steganography is used for secrecy.

3. What is unique about messages transmitted using steganography?

   Most of the time messages sent by steganography is hidden/embedded in an ordinary looking file

   This results in a secrecy of real message while the file looks completely normal to an ordinary person.

   Only specialized tools are able to extract the message which is why it is hard to detect and expect in a file like image, video, audio or text file.

4. What are the basic steganographic concepts or inputs and outputs?

   To perform steganography (Input) we require 3 things: 1) Object like image, video, etc. 2) Message to be hidden 3) Key: use to safeguard the secret message (even if someone finds that file is Stego file they can' t access the message unless they get the key.)

   For extracting message (Output) we need the stego file and tool to extract it and key

References

Alun Jones. (2020, June 24). *Revisiting NTFS Alternate Data Streams*. MSMVPs Blog. https://blogs.msmvps.com/alunj/2020/06/24/revisiting-ntfs-alternate-data-streams/

Microsoft. (n.d.). *File Streams*. Microsoft Learn. https://docs.microsoft.com/en-us/windows/win32/fileio/file-streams